

Erik Mischiatti

Robotics / UAV Test Engineer | Mechatronics Engineer
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Portfolio | LinkedIn | GitHub

Professional Summary

Mechatronics engineer focused on robotic and UAV systems from integration to validation. Combines robotics software, embedded prototyping, and simulation-based testing to turn autonomy logic into reliable field behavior.

Technical Skills

Robotics / UAVs: ROS1, BehaviorTree.CPP, MRS UAV System, Gazebo, RViz, TF, mission-level services, autonomous mission logic
Programming / Systems: C++, Python, MATLAB, Bash, Linux, Git, Docker, integration testing, simulation campaigns, failure-mode validation
Embedded / Hardware: UWB communication, PWM, ESC/motor modules, PCB design, RTK GNSS, LoRa, sensor integration
Perception / Prototyping: OpenCV, ArUco, RealSense D435i, RGB-D cameras, Fusion 360, SolidWorks, 3D printing

Professional Experience

Czech Technical University in Prague, Dept. of Cybernetics

Prague, Czech Republic

Test Engineer – EAGLE.ONE Drone Hunting System

Oct 2025 – Present

- **Mission Orchestration:** Develop and test BehaviorTree.CPP logic for preflight, takeoff, pre-intercept, interceptor, post-intercept, landing, and finalization phases.
- **Simulation-Based Validation:** Execute MRS UAV System campaigns to verify safety-critical mission transitions, readiness gates, edge cases, and failure handling.
- **Integration Debugging:** Debug issues across mission manager, interceptor supervision, MQTT/UI interfaces, payload services, and simulation environments.
- **RTK GNSS Deployment:** Design and deploy a base-station setup for high-accuracy UAV positioning, including hardware selection, enclosure design, and field deployment.

Technical University of Vienna, Institute of Autonomous Systems

Vienna, Austria

Master Thesis Researcher

Apr 2024 – Oct 2024

- **Robotic Manipulation:** Developed a ROS-based cable-insertion framework using Franka Emika Panda, kinesthetic teaching, learning from demonstration, and invariant trajectory representation.
- **Perception Integration:** Integrated RealSense D435i sensing, OpenCV/ArUco pose estimation, ROS TF, and Behavior Tree sequencing for modular assembly scenarios.

Selected Projects

M.I.R.A. – Modular Interactive Robotic Agent

Portfolio / GitHub

Software-first embodied assistant prototype

- Built a Python/PySide6 desktop assistant with animated state feedback, chat interaction, session memory, modular cognition, and constrained local actions for future robotic embodiment.

Modular UWB Drone Communication System

Portfolio

Embedded wireless propulsion-unit communication for UAVs

- Designed UWB links between a central controller and distributed ESC/motor modules; implemented PWM capture, synchronization, ESC command forwarding, and custom PCB integration.

Green Vigilance – UAV/UGV Precision Agriculture Simulation

Portfolio / GitHub

Python simulation for early crop disease detection

- Built a reproducible multi-robot simulation with YAML scenarios, UAV dynamics, localization estimation, UGV target assignment, disease heatmaps, metrics, and comparison reports.

Education

University of Trento, M.Sc. Mechatronics Engineering – thesis at TU Wien

2022 – 2024

Linnaeus University, Erasmus exchange, robotics and automation – Vaxjo, Sweden

2023

University of Padova, B.Sc. Mechatronics Engineering – Vicenza / Padova, Italy

2018 – 2021

Publications, Awards, and Languages

IEEE CASE 2024: Modular Drone / UWB Communication project, IEEE 20th International Conference on Automation Science and Engineering. | **DigiEduHack 2021:** Global Award Winner, Hachi Team. | **Languages:** Italian native, English B2/C1, German A1, Swedish A1.